TERRENCE LEI

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EDUCATION

University of California - Santa Barbara **August 2023 - June 2027**
Bachelor of Science (BS), Mechanical Engineering
- Specialization in Dynamic Systems, Control, and Robotics
- Relevant Coursework: Controls, Mechatronics, Fluids, Thermodynamics, Machine Learning, Circuits, Vibrations

University of California - Santa Barbara **January 2024 - June 2027**
Bachelor of Science (BS), Statistics and Data Science

PROFESSIONAL EXPERIENCE

Applied Materials **June 2026 - September 2026**
Technical Project Management Intern *Santa Clara, CA, USA*
- Designed a patent pending motorized actuator mechanism capable of two distinct directions of movement from a single motor source
- Positioned as an upgrade solution with extended capabilities including sensor data, automated optimizations, and predictive insights to extend the operating lifetime for 1000+ tools worldwide
- Engaged with suppliers to secure prototype parts adhering to a strict timeline in a cost effective manner

University of California - Santa Barbara **January 2026 - Present**
Undergraduate Research Assistant *Santa Barbara, CA, USA*
- Developing automated system to monitor workbench status, robotic movements and report results including a custom camera sensor suite
- Training a robotic arm for automated testing sequences, with the potential to support 50k material tests a day
- Designed data flow from raw test results managed with SQL to actionable insights

Northrop Grumman **June 2025 - August 2025**
Mechanical Engineering Intern *Goleta, CA, USA*
- Designed and conducted component and process experiments for assembly failure analysis, using regression and classification methods to identify the root causes and reduce production times and material costs
- Engineered a solar array test vehicle used to develop wiring techniques, optimizing assembly ergonomics to reduce lead times
- Worked on tolerance analysis for legacy flight components and released new 2D GD&T drawings

IGEL Technology **June 2023 - Present**
Financial Analyst Intern *Remote*
- Building Excel-based financial models to calculate revenue recognition differences, pricing allocations and adjustment forecasts, creating a more efficient process that reduced analysis times from 3 months to 1 week

PROJECTS & OUTSIDE EXPERIENCE

Autonomous Shopping Cart **April 2026 - June 2026**
Undergraduate Student *Santa Barbara, CA, USA*
- Engineered an autonomous shopping cart as an integrated consumer product, translating shopper behavior into requirements for form factor, maneuverability, safety, controls, and an intuitive user experience
- Integrated a complex electromechanical system spanning motors and encoders, power electronics, embedded control, onboard compute, and machine vision, balancing packaging, performance, reliability, power, and cost
- Iteratively tested the interaction experience including following, obstacle avoidance, aisle navigation, target reacquisition, and visual feedback

Gaucho Rocket Project **September 2024 - June 2026**
Vice President *Santa Barbara, CA, USA*
- Integrated structures, propulsion, actuators, sensors, custom PCBs, power electronics, and controls into tightly packaged flight hardware
- Performed FEA and structural optimization of components; iterating geometry, load paths, and material selection to minimize mass while meeting stiffness, strength, and factor of safety requirements
- Designed for manufacturing across CNC machining, additive manufacturing, sheet metal, and composite processes
- Organized and led design reviews in partnership with faculty members and industry sponsors

SKILLS

Programming: C/C++, Java, Python, SQL, HTML/CSS, R, MATLAB, LaTex, VS Code, PyCharm, ROS, Pandas, Pytorch
Technical: SolidWorks, Fusion 360, Ansys CFX, MSC Adams, Creo, Windchill, Simulink, Excel, GD&T, LabView, NX, Siemens Teamcenter